

# PSC Functional Test Report

| PSC Functional Test Results           |                 |
|---------------------------------------|-----------------|
| Power Supply Controller Configuration |                 |
| Description                           | PSC-4CH-HSF-EIC |
| Serial Number                         | 0001            |
| Number of Channels                    | 4               |
| Resolution                            | HS (20bit)      |
| Bandwidth                             | F               |
| Channel 1 Polarity                    | Bipolar         |
| Channel 2 Polarity                    | Bipolar         |
| Channel 3 Polarity                    | Bipolar         |
| Channel 4 Polarity                    | Bipolar         |

## FOFB TX test (Bandwidth-Mode = Fast)

| PV                | Value              | Pass? |
|-------------------|--------------------|-------|
| lab{2}Chan1:DAC-I | 11.500301361083984 | PASS  |
| lab{2}Chan2:DAC-I | 11.500070571899414 | PASS  |
| lab{2}Chan3:DAC-I | 11.50058650970459  | PASS  |
| lab{2}Chan4:DAC-I | 11.500749588012695 | PASS  |

## UDP RX Packet Test

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14:49:17.193842 IP (tos 0x0, ttl 64, id 14145, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.51845 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
14:49:17.194500 IP (tos 0x0, ttl 64, id 14146, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.39279 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
14:49:17.205308 IP (tos 0x0, ttl 64, id 14151, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.35592 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
14:49:17.206124 IP (tos 0x0, ttl 64, id 14152, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.42118 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
14:49:17.217025 IP (tos 0x0, ttl 64, id 14159, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.59623 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
14:49:17.217908 IP (tos 0x0, ttl 64, id 14160, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.56165 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
14:49:17.228816 IP (tos 0x0, ttl 64, id 14171, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.49407 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
14:49:17.229579 IP (tos 0x0, ttl 64, id 14172, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.43055 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
14:49:17.240338 IP (tos 0x0, ttl 64, id 14182, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.58561 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
14:49:17.241085 IP (tos 0x0, ttl 64, id 14183, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.53181 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
14:49:17.252203 IP (tos 0x0, ttl 64, id 14192, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.51105 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
14:49:17.252926 IP (tos 0x0, ttl 64, id 14193, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.54549 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
14:49:17.264250 IP (tos 0x0, ttl 64, id 14201, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.39520 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
14:49:17.264801 IP (tos 0x0, ttl 64, id 14202, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.39352 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
14:49:17.276130 IP (tos 0x0, ttl 64, id 14211, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.53274 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
14:49:17.276450 IP (tos 0x0, ttl 64, id 14212, offset 0, flags [DF], proto UDP (17), length 64)
  10.69.26.1.57279 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
14:49:17.288053 IP (tos 0x0, ttl 64, id 14215, offset 0, flags [DF], proto UDP (17), length 64)
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10.69.26.1.53980 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
14:49:17.288056 IP (tos 0x0, ttl 64, id 14216, offset 0, flags [DF], proto UDP (17), length 64)
10.69.26.1.44709 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
14:49:17.299828 IP (tos 0x0, ttl 64, id 14220, offset 0, flags [DF], proto UDP (17), length 64)
10.69.26.1.52280 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
14:49:17.299830 IP (tos 0x0, ttl 64, id 14221, offset 0, flags [DF], proto UDP (17), length 64)
10.69.26.1.46414 > 10.69.26.55.12345: [udp sum ok] UDP, length 36
```

(Full tcpdump log saved to /home/pstester/NSLS-II\_PSCs/NEXT-II/NSLS2\_zPSC\_Calibration\_and\_Test/test\_data/raw\_logs/4ch\_HSF\_SN0001\_RawData\_05-19-26\_14-49/tcpdump\_full.log)

| Test        | Result |
|-------------|--------|
| UDP RX FOFB | PASS   |